NAMAA Community at DialectSentEval 2026 Subtask 1: Dialect-Aware Arabic Sentiment Classification with Pretrained Transformer Models

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Abstract

Arabic dialect sentiment analysis is challenging because regional varieties show considerable lexical, morphological, syntactical, and orthographic variation. This paper presents a transformer-based system developed for DialectSentEval 2026 Subtask 1, a three-class sentiment classification task covering Moroccan, Egyptian, Jordanian, and Saudi Arabic. The system independently fine-tunes three pretrained Arabic language models: AraBERTv2, MARBERTv2, and CAMeLBERT, using a common training protocol. The implementation uses an 80/20 stratified train-validation split, a maximum sequence length of 128, a batch size of 16, a learning rate of 2×10^{-5}, and focal loss with $\gamma = 2$=2. Model selection was conducted based on validation macro F1. Using the published labeled test set, MARBERTv2 achieves the strongest independent result with 94.07% macro F1, followed by CAMeLBERT at 81.96% and AraBERTv2 at 79.77%. The results demonstrate the importance of Arabic pretraining and dialect-oriented representations and establish MARBERTv2 as the strongest standalone backbone in the current system.

1 Introduction

Arabic dialect sentiment analysis remains challenging because of regional variation, morphological richness, orthographic inconsistency, and frequent divergence from Modern Standard Arabic (MSA) (Nwesri et al., 2025). These characteristics are evident in user-generated content, where dialectal vocabulary, informal spelling, code-switching, and non-standard forms are common. Consequently, sentiment classifiers trained primarily on standardized Arabic may not transfer uniformly across dialects.

DialectSentEval 2026 Subtask 1 (Ezzini et al., 2026a) provides a controlled benchmark for this setting. The task requires sentence-level prediction of negative, neutral, or positive sentiment across Moroccan, Egyptian, Jordanian, and Saudi Arabic. The shared-task resource is derived from hotel reviews in the ADOR dataset (Alharbi et al., 2025c) and contains parallel sentences across dialects, with balanced sentiment representation (Ezzini et al., 2026b).

The official benchmark uses an 80/20 train-test split based on sentence identifiers so parallel sentences remain in the same partition, reducing leakage risk. The official evaluation uses macro F1, which assigns equal weight to the three sentiment classes. The system's main objective is to enable a reproducible comparison of three Arabic pretrained encoders under a common fine-tuning regime.

2 Related Work

Arabic-specific pretraining has consistently improved downstream Arabic NLP performance. AraBERT (Antoun et al., 2020) demonstrated the effectiveness of language-specific BERT pretraining across several tasks, including sentiment analysis, and motivated subsequent Arabic-focused encoders. ARBERT and MARBERT (Abdul-Mageed et al., 2021) extended this direction by targeting the diversity of Arabic varieties, with MARBERT emphasizing dialectal Arabic and achieving strong results on the multi-dialect ARLUE benchmark.

The importance of pretraining-data composition was further examined by (Inoue et al., 2021), who showed that similarity between the pretraining language variety and the downstream data can be more important than corpus size alone. More recently, AraModernBERT (Elshehy et al., 2026) introduced transtokenized initialization and long-context Arabic encoder modeling, further demonstrating the value of specialized Arabic representations.

These findings are consistent with broader comparative studies of Arabic NLP architectures. (Mohamed and Alosman, 2025) reported that pretrained transformer models outperform earlier RNN- and

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CNN-based approaches across multiple Arabic NLP tasks, including sentiment analysis and dialect identification.Recent work has further emphasized the importance of evaluating sentiment analysis methods specifically in dialectal Arabic settings. Alharbi et al. (Alharbi et al., 2025a) examined large language models for Arabic dialect sentiment analysis, while the AHASIS shared task provided a dedicated evaluation setting for sentiment analysis across Arabic dialects (Alharbi et al., 2025b). Together, this prior work motivates our comparison of Arabic encoders with different pretraining characteristics for multi-dialect sentiment classification.

3 Data Description

3.1 Dataset Statistics

The experiments used the supplied DialectSentEval Subtask 1 training and test files. Table 1 shows the distribution of sentiment labels across the four Arabic dialects. The training file contains 1,731 labeled Arabic sentences, while the test file contains 312 instances. The training records include ID, sentence, sentiment, and dialect fields. Sentiment is represented as a three-class problem: Class 0 (negative), Class 1 (neutral), and Class 2 (positive). The data cover four Arabic varieties: Moroccan Darija, Egyptian, Jordanian, and Saudi Arabic.

The training set is almost perfectly balanced across dialects, as shown in Table 1. Moroccan Darija, Saudi, and Jordanian each contribute 433 instances, whereas Egyptian contributes 432 instances. Thus, each dialect represents approximately one quarter of the training corpus. By contrast, the sentiment distribution is moderately imbalanced: negative examples account for 657 instances (37.95%), positive examples for 609 (35.18%), and neutral examples for 465 (26.86%). The smaller neutral class is particularly relevant to model selection because sentences with weakly polarized sentiment can be more difficult to distinguish from both positive and negative examples.

Table 1: Distribution of Sentiment Labels Across Arabic Dialects

Dialect	**Neg.**	**Neu.**	**Pos.**	**Total**
Darija	166	115	152	433
Egyptian	159	120	153	432
Jordanian	166	115	152	433
Saudi	166	115	152	433
Total	**657**	**465**	**609**	**1,731**

3.2 Training Validation Protocol

For model development, the 1,731 labelled training instances are divided using an 80/20 stratified split with random seed 42. Stratification was performed on the sentiment label to preserve the relative proportions of the three classes in the training and validation subsets. This produces approximately 1,385 training instances and 346 validation instances. The 312-instance test set is kept separate from this local validation process and is not used for model selection. This distinction is important when interpreting the reported results. The validation set monitors convergence and selects the best checkpoint, whereas the supplied test set represents the held-out evaluation data. The experiments therefore avoid using test performance to determine model configuration or checkpoint selection.

3.3 Input Representation and Preprocessing

The sentence field is used as the model input without additional handcrafted linguistic features. Each sentence is tokenised using the model-specific tokeniser, padded to a maximum sequence length of 128 tokens, and truncated when necessary. No external sentiment lexicon, stemming, lemmatisation, or manually engineered dialect features are introduced. This design isolates the contribution of the pretrained Arabic transformer representations.

4 Model Selection and Experimental Configuration

4.1 Model Selection Strategy

Three Arabic pretrained transformer backbones are evaluated: AraBERTv2, MARBERTv2, and CAMeLBERT. They were selected to provide complementary Arabic representations and, in particular, to contrast a general Arabic pretrained encoder with models developed to better represent Arabic dialectal variation. All three models are fine-tuned under an identical protocol so that differences in performance can be attributed primarily to the pretrained representations rather than model-specific optimization choices.

4.2 AraBERTv2

AraBERTv2 represents the AraBERT family of Arabic-specific pretrained language models. AraBERT demonstrated the effectiveness of language-specific BERT pretraining for Arabic natural language understanding and downstream sentiment-related tasks (Antoun et al., 2020). In this study, AraBERTv2 provides a strong general Arabic baseline against which more dialect-oriented representations can be compared.

4.3 MARABERTv2

MARBERTv2 is particularly relevant to the present task because the MARBERT family was developed to address the diversity of Arabic language varieties. (Abdul-Mageed et al., 2021) introduced MARBERT together with ARBERT and evaluated the models on ARLUE, a multi-dialect Arabic language-understanding benchmark. Their findings demonstrated strong performance across diverse Arabic NLP tasks. Moreover, (Inoue et al., 2021) showed that the proximity between the pretraining language variant and downstream data can be more important than pretraining-data size. These findings motivate MARBERTv2 as the principal dialect-oriented candidate in the current comparison.

4.4 CAMeLBERT

CAMeLBERT is included as a complementary Arabic pretrained representation from the CAMeL-Lab model family (Inoue et al., 2021). Its inclusion provides an additional Arabic-specific baseline and helps distinguish the general benefits of Arabic pretraining from the more dialect-focused characteristics of MARBERTv2.

4.5 Common Fine-Tuning Configuration

All three backbones use the same fine-tuning configuration: a maximum sequence length of 128 tokens, training and evaluation batch sizes of 16, a learning rate of 2×10^{-5}, 10 training epochs, focal loss with $\gamma = 2$, and random seed 42. Evaluation is performed after each epoch, and the checkpoint with the highest validation Macro-F1 is retained.

To emphasize difficult training examples, the system uses focal loss. Given the cross-entropy loss

$$p_t = \exp(-CE) \tag{1}$$

, and focusing parameter $\gamma = 2$, the loss is defined as

$$\mathcal{L}_{\text{focal}} = (1 - p_t)^{\gamma} CE \tag{2}$$

. The same objective is applied to all three models, ensuring that the comparison does not confound backbone selection with different optimization objectives.

Table 2: Performance comparison of pretrained Arabic language models.

Model	Epoch	Accuracy	Macro P	Macro R	macro F1
AraBERTv2	7	0.9164	0.9092	0.9095	0.9092
MARBERTv2	6	0.9568	0.9531	0.9547	0.9539
CAMeLBERT	9	0.8991	0.8918	0.8935	0.8923

5 Results

5.1 Validation Performance

Table 1 summarizes the strongest validation checkpoint observed during the training. The three models exhibit different convergence profiles. MARBERTv2 reaches its maximum macro F1 at epoch 6, whereas AraBERTv2 peaks at epoch 7 and CAMeLBERT at epoch 9. The distinction between best-epoch validation performance and final-epoch performance is important because the checkpoint-selection criterion is validation macro F1.

5.2 Class-Level Behavior

The per-class validation results reveal a consistent difficulty pattern. For MARBERTv2 at its best epoch, Class 0 reaches F1=0.9771, Class 1 reaches F1=0.9255, and Class 2 reaches F1=0.9590. AraBERTv2 obtains Class 0, Class 1, and Class 2 F1 values of 0.9699, 0.8449, and 0.9129 at its best epoch, respectively. CAMeLBERT obtains 0.9535, 0.8211, and 0.9024 at its best epoch. Thus, the neutral class (Class 1) is the most challenging class for all three models, whereas negative sentiment (Class 0) is consistently the easiest.

As shown in Table 2, MARBERTv2 not only achieves the highest macro F1 but also maintains comparatively strong performance on the neutral class. This behavior suggests that its pretrained representations provide more discriminative contextual information for cases that are less lexically polarized.

5.3 Dialect-wise System Performance

To assess robustness across Arabic varieties, we evaluated predictions on the 1,731 labeled training instances, which are nearly evenly distributed

Dialect	Model	Accuracy	Macro-P	Macro-R	Macro-F1
Moroccan	AraBERTv2	94.23	93.69	94.01	93.82
	MARBERTv2	**97.46**	**97.23**	**97.52**	**97.36**
	CAMeLBERT	97.00	96.98	96.83	96.90
Egyptian	AraBERTv2	94.68	94.62	94.05	94.26
	MARBERTv2	**96.99**	**96.94**	**96.76**	**96.84**
	CAMeLBERT	95.60	95.45	95.28	95.36
Jordanian	AraBERTv2	97.46	97.20	97.27	97.23
	MARBERTv2	**99.54**	**99.49**	**99.49**	**99.49**
	CAMeLBERT	98.38	98.35	98.22	98.28
Saudi	AraBERTv2	96.30	96.36	95.64	95.94
	MARBERTv2	**98.61**	**98.57**	**98.40**	**98.48**
	CAMeLBERT	98.38	98.46	98.04	98.23

Table 3: Performance of the three encoder models across the four Arabic dialects. Results are reported as percentages. Bold indicates the best score for each dialect and metric.

across Moroccan Darija (433), Egyptian (432), Jordanian (433), and Saudi (433). This balance enables direct comparison across dialects without major sample-size effects.

As shown in Table 3, MARBERTv2 consistently achieves the best Macro-F1 across all dialects, with 99.49% on Jordanian, 98.48% on Saudi, 97.36% on Moroccan Darija, and 96.84% on Egyptian Arabic. AraBERTv2 shows greater variation, ranging from 93.82% on Moroccan Darija to 97.23% on Jordanian, while CAMeLBERT ranges from 95.36% on Egyptian to 98.28% on Jordanian.

Overall, Jordanian Arabic is the easiest dialect across all three models, while Moroccan Darija and Egyptian Arabic are comparatively more challenging. MARBERTv2's consistent advantage suggests that its dialect-oriented pretraining provides stronger representations for the linguistic variability in this task, which is also reflected in the held-out evaluation, where it achieves 94.07% Macro-F1 compared with 81.96% for CAMeLBERT and 79.77% for AraBERTv2.

5.4 Convergence and Generalization

AraBERTv2 improves rapidly during the first three epochs, reaching macro F1=0.9038 at epoch 3, followed by smaller fluctuations and a maximum of 0.9092 at epoch 7. MARBERTv2 improves from 0.7112 at epoch 1 to 0.9441 at epoch 5 and reaches 0.9539 at epoch 6, after which its validation performance stabilizes near 0.951. CAMeLBERT exhibits a slower but sustained improvement, increasing from 0.6601 at epoch 1 to 0.8923 at epoch 9 before a small decline at epoch 10. These trajectories justify retaining the best checkpoint rather than automatically using the final epoch.

6 Discussion

The experimental results indicated that pretrained representation choice is a major determinant of performance in this multi-dialect setting. MARBERTv2 consistently outperforms AraBERTv2 and CAMeLBERT under the same optimization system. This result is consistent with prior findings that dialectal data exposure and pretraining variant proximity matter for downstream Arabic NLP tasks. It also aligns with the literature review's observation that MARBERTv2 has performed strongly on dialectal sentiment benchmarks.

The neutral-class weakness deserves particular attention. Although the benchmark is balanced at the dataset level, neutral sentiment is semantically less polarized than negative or positive sentiment and may contain lexical cues that overlap with both classes. The substantially higher neutral F1 obtained by MARBERTv2 suggests that dialect-oriented pretraining may help capture contextual distinctions that are difficult for more general Arabic representations.

7 Conclusion

This study presented a transformer-based system for DialectSentEval 2026 Subtask 1 and evaluated AraBERTv2, MARBERTv2, and CAMeLBERT independently under a shared training protocol. The system uses stratified validation, fixed random seeds, and validation macro-F1 checkpoint selection to promote the best model. Across the validation runs, MARBERTv2 achieved the highest performance with a macro F1 of 0.9539 using only

6 epochs, and then AraBERTv2 and CAMeLBERT achieved macro F1 of 0.9092 and 0.8923, respectively. Additionally, MARBERTv2 also showed the strongest class-balanced behavior, particularly for the neutral class.

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